COMPUTER SCIENCE AND ENGINEERING
INTERNET OF THINGS (IoT)
2025 - 2026

Challenge #2
Packet Sniffing

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Part 2 - EXERCISE

A smart building system includes:

- A battery-powered temperature sensor that transmits a measurement every 2 minutes.
- A battery-powered actuator (fan controller) that receives temperature data and decides whether to turn ON/OFF.
- A gateway (mains-powered) that only supports MQTT.

The sensor and actuator can communicate either:

- Directly using CoAP
- Through the gateway using MQTT.

Consider the following message sizes (in bytes), which already include header and payload size:

COAP		MQTT	
CON PUT Request	70 B	Subscribe	65 B
PUT ACK	15 B	Sub Ack	55 B
		Publish	75 B
		Pub Ack	50 B
		Connect/Disconnect	60 B
		Connect Ack	50 B

Assume that:

1. Transmit and Receive cost per bit are $E_{TX} = 50nJ/bit,\ E_{RX} = 58nJ/bit$.
2. CoAP communication uses Confirmable (CON) messages.
3. MQTT communication uses QoS 1 (for both actuator and sensor sides).
4. Ideal communication (no losses).
5. When retransmission is required, $RTO = 30s$ and $MAX_{TRANSSIONS} = 3$.
6. Devices start disconnected.
7. MQTT connections are closed gracefully during sleep periods (with non-acknowledged DISCONNECT).
8. Only the energy consumed by the sensor and actuator must be considered (ignore gateway consumptions).

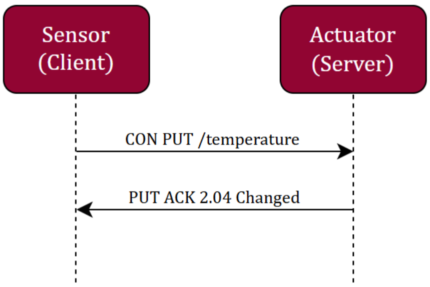
EQ1:

Compute the total communication energy consumed by the battery-powered sensor and actuator over 1 hour in:

a) CoAP (direct communication).

b) MQTT (via the gateway).

Provide numerical results in μJ



a) CoAP-based communication

For transmit:

- $E_{TX\_MSG} = Message\ Size\ (Bytes) \times 8 \times 50nJ/bit$

For receive:

- $E_{RX\_MSG} = Message\ Size\ (Bytes) \times 8 \times 58nJ/bit$

Sensor node energy:

- $E_{TX\_PUT\_Request} = 70B \times 8 \times 50nJ/bit = 28000nJ = 28\mu J$
- $E_{RX\_PUT\_ACK} = 15B \times 8 \times 58nJ/bit = 6960nJ = 6,96\mu J$
- $E_{CoAP\_Sensor} = (28 + 6,96)\mu J = 34,96\mu J$

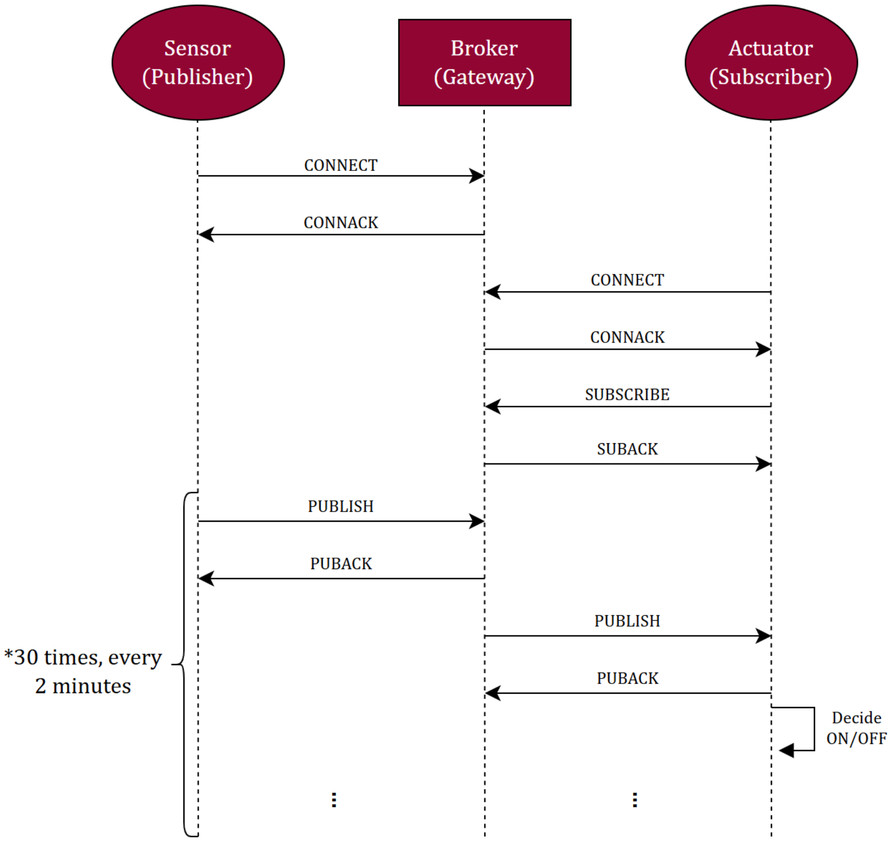
Actuator node energy:

- $E_{RX\_PUT\_Request} = 70B \times 8 \times 58nJ/bit = 32480nJ = 32,48\mu J$
- $E_{TX\_PUT\_ACK} = 15B \times 8 \times 50nJ/bit = 6000nJ = 6\mu J$
- $E_{CoAP\_Actuator} = (32.48 + 6)\mu J = 38,48\mu J$

Total energy consumption of the system by the battery-powered sensor and actuator over 1 hour:

- $1h = 60min \Rightarrow \frac{60min}{2min} = 30\ times$
- $E_{CoAP\_Tot} = 30 \times (E_{CoAP\_Sensor} + E_{CoAP\_Actuator}) =$

$$= 30 \times (34,96 + 38,48)\mu J = 2203,2\mu J$$



b) MQTT-based communication (QoS 1)

Sensor node energy

- $E_{CONNECT} = 60B \times 8 \times 50nJ/bit = 24000nJ = 24\mu J$
- $E_{CONNACK} = 50B \times 8 \times 58nJ/bit = 23200nJ = 23,2\mu J$
- $E_{PUBLISH} = 75B \times 8 \times 50nJ/bit = 30000nJ = 30\mu J$
- $E_{PUBACK} = 50B \times 8 \times 58nJ/bit = 23200nJ = 23,2\mu J$
- $E_{QMTT\_Sensor} = [24 + 23,2 + 30 \times (30 + 23,2)]\ \mu J = 1643,2\ \mu J$

Actuator node energy

- $E_{CONNECT} = 60B \times 8 \times 50nJ/bit = 24000nJ = 24\mu J$
- $E_{CONNACK} = 50B \times 8 \times 58nJ/bit = 23200nJ = 23,2\mu J$
- $E_{SUBSCRIBE} = 65B \times 8 \times 50nJ/bit = 26000nJ = 26\mu J$
- $E_{SUBACK} = 55B \times 8 \times 58nJ/bit = 25520nJ = 25,52\mu J$
- $E_{PUBLISH} = 75B \times 8 \times 58nJ/bit = 34800nJ = 34,8\mu J$
- $E_{PUBACK} = 50B \times 8 \times 50nJ/bit = 20000nJ = 20\mu J$
- $E_{MQTT\_Actuator} = [24 + 23,2 + 26 + 25,52 + 30 \times (34,8 + 20)]\ \mu J = 1742,72\ \mu J$

Total energy consumption of the system by the battery-powered sensor and actuator over 1 hour:

- $1h = 60min \Rightarrow \frac{60min}{2min} = 30\ times$
- $E_{Tot\_MQTT} = E_{QMTT\_Sensor} + E_{MQTT\_Actuator} = (1643,2 + 1742,72)\ \mu J = 3385,92\ \mu J$

EQ2:
Assume now that the actuator wakes up only once every 30 minutes.
Which protocol would you choose for long-term operation?
Justify your answer (max 300 characters), explicitly stating any protocol parameter required (e.g., clean session, retain, observe, block-transfer modes etc...) and performance metric you optimize.

MQTT can minimize energy consumption because, instead of CoAP, the actuator does not need to wait up to 2 minutes for the next sensor transmission; with retained messages, it can retrieve the latest value immediately and return to sleep sooner.

More detailed answer:
In our opinion, MQTT is preferable to CoAP for long-term operation.
With a persistent session, the broker can store subscription state and queue QoS 1 messages while the actuator is asleep, allowing missed updates to be delivered after reconnection. In contrast, CoAP requires both nodes to be simultaneously online, so messages sent every 2 minutes from the sensor would be lost, and the actuator may need to wait up to 2 minutes after waking up to receive a new update.
Actually, the persistent session is not the most efficient solution because it delivers a lot of redundant updates. Therefore, from our perspective, using retained messages is more energy-efficient, since the actuator only needs the latest value to make a decision.
The performance metric optimized is the minimization of energy consumption, thus maximizing battery lifetime.